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SARDAR PATEL UNIVERSITY

S.Y.B.C.A. SEMESTER – III (CBCS) EXAMINATIONS

US03CBCA03 : ADVANCED DATA & FILE STRUCTURES

Friday, 17 January, 2020

Time: 10:00 AM to 1:00 PM

Max. Marks: 70

Q-1 Multiple Choice Questions. **[10]**

1. An array is called finite data structure because____________________.
- A. It contains infinite number of elements.
- B. It contains unlimited elements.
- C. It contains limited number of elements.
- D. It does not contain limited number of elements.

2. ______ is a sequence of consecutive edges from the source node to destination node.
- A. Path
- B. Edge
- C. Leaf
- D. Height

3. ____________ contains maximum possible number of nodes in all level except possibly the last level.
- A. Skew binary tree
- B. Full binary tree
- C. Complete binary tree
- D. None of these

4. In linear representation of binary tree, for any node with index i, where 1 < i <= n, the right child of that node is located at __________ index.
- A. 2 * i
- B. 2 * i + 1
- C. 2 * i - 1
- D. i / 2

5. A node whose outdegree is 0 is called __________.
- A. Source node
- B. Sink node
- C. Self loop
- D. Single node

6. During each pass of the selection sort, no more than one interchange is required, the maximum number of interchanges for the sort is ____________.
- A. n - 1
- B. n
- C. n + 1
- D. n / 2

7. ______________technique requires an ordered table to search a particular record in the table.
- A. Sequential search
- B. Binary search
- C. Sorting
- D. None of these

8. ____________ processing is the accessing of records, one after the other, according to the physical order in which they appear in the file.
- A. Serial
- B. Direct
- C. Sequential
- D. None of these

9. The lowest level of index is ____________.
- A. Master index
- B. Prime index
- C. Index area
- D. Track index

10. A ______________ which defines a mapping from key space to the address space.
- A. Deblocking
- B. Blocking
- C. Hashing Function
- D. None of these

Q-2 Attempt any ten. **[20]**

1. List any 4 applications of tree.
2. A two dimensional array defines as a [-3…4, 1…6] requires 2 bytes of storage space per each element. If the array is stored in row major form then calculate the address of element at location a [1, 3]. Base address is 251.
3. Define: Height and Siblings
4. Construct binary tree from following traversal.

Inorder	4	2	5	1	6	7	3	8
Postorder	4	5	2	6	7	8	3	1

5. Define: Strongly connected graph and Loop
6. Draw binary tree for (A / (B + C) * D) - (E * F + G / H)
7. Give any 4 applications of searching.
8. Sort the array of 5 elements 37, 4, 51, 23, 12 using Selection sort.
9. Differentiate: Searching and Sorting.
10. Why collision resolution technique is used? List out different classes of collision resolution techniques.
11. Define Cross reference table with example.
12. Define: Record, Synonym

Q-3

(A) Write note on: Applications of array. **[5]**

(B) Explain address calculation of 2-D array element with example. **[5]**

OR

Q-3

(A) Explain address calculation of 1-D array element with example. **[5]**

(B) Write note on: Sparse Matrix **[5]**

Q-4

(A) Explain linear representation method for implementing Binary tree. **[5]**

(B) Write an algorithm for Inorder traversal of a binary tree. **[5]**

OR

Q-4

(A) Explain deletion in a binary tree (all conditions) with example. **[5]**

(B) Explain Threaded binary tree with advantages and disadvantages. **[5]**

Q-5

(A) Write down the algorithm of Bubble sort. **[5]**

(B) Write down the algorithm of Binary search. **[5]**

OR

Q-5

(A) Write note on: Applications of sorting. **[5]**

(B) Write down the algorithm of Sequential Search. **[5]**

Q-6 Write a detail note on structure and processing of Sequential file organization. **[10]**

OR

Q-6 Write a detail note on structure and processing of Indexed sequential file organization supported by CDC. **[10]**

☺ ☺ ☺ ☺ ☺ **Good Luck** ☺ ☺ ☺ ☺ ☺